

Wealth Gradient in Drinking Water Quality in African Countries

This paper examines the wealth gradient in having improved drinking water at point of use and compares that gradient between rainy and dry seasons. Pooling nationally representative survey data from 14 African countries, I show that households in the richest quintile are associated with a 21.2 percentage point increase in the probability of having improved drinking water at point of use compared to the poorest households in the dry season and I show a differential wealth gradient in the rainy season. However, due to substantial variation across countries, this estimate is close to zero and is statistically insignificant when including country fixed effects. I study a household's endogenous decision to treat the water or choose a better source to examine the mechanisms driving the wealth gradient: I show the wealth gradient is driven by households in the fourth and fifth quintile being more likely to use better water sources compared to households in the poorer wealth quintiles, consistent with and without the country fixed effects. In contrast, the estimate for the decision to treat the water is not consistent with and without the country fixed effects.

CHAPTER ONE

Introduction

Since 2015, the number of people without safely managed drinking water in sub-Saharan Africa has increased from 703 to 766 million (United Nations, 2021). The wealth gradient in health—indicating poorer health among less affluent individuals—has long intrigued health policy and social science discussions (Kawachi, Kennedy, Lochner, & Prothrow-Stith, 1997; Kennedy, Kawachi, & Prothrow-Stith, 1996; Wilkinson & Pickett, 2008). Wealthier people on average appear to have better health. Theory suggests that the same might apply to drinking water quality and that wealthier people, on average, might have better drinking water. Yet, the wealth gradient in water quality has scarcely been explored in economics.¹ This paper looks at the wealth gradient in water quality at the point of use, through improved sources or water treatment.

In order to analyze drinking water quality, the primary variable of interest I use is a measure of having clean drinking water. I define that in the following way: I construct a binary variable, “Improved Water” that takes the value of “1” if a household has better water at the point of consumption than point of use. I use this variable as my measure of clean drinking water because studies typically do not take into account the degradation in drinking water quality that happens from the source to the point of consumption as Jessoe (2013) states. A primary

¹ The Economics literature that has examined water supply has mainly focused on the following research strands: the impact of adequate water supply on waterborne diseases (Zwane & Kremer, 2007), child mortality (Gamper-Rabindran, Khan, & Timmins, 2010), educational attainment (Gould, Lavy, & Paserman, 2011), and women’s empowerment (Ivens, 2008), as well as its connection to corruption (Anbarci, Escaleras, & Register, 2009) and different systems of government (Deacon, 2009).

contribution of this paper is to use this degradation in water quality as a measure of having clean drinking water.

I analyze data from 27,408 households from 14 African countries, drawn from the Multiple Indicator Cluster Survey (MICS) datasets from the sixth round (MICS6). I observe a wealth gradient in the probability of having improved drinking water in the dry season. Being in the poorest wealth quintile is associated with a 22.1 percentage point decrease in the probability of having improved water at point of use compared to households in the richest wealth quintile in the dry season.

A strand of literature has found the adverse effect rainy seasons have on water quality. Systematic reviews suggest fecal contamination in drinking water sources follow a seasonal trend of greater contamination during the rainy season.² The greater contamination disproportionately affects poorer households (WHO 2019). The report only focuses on how the poor do not have access to clean water infrastructure and claims the lack of infrastructure might affect them negatively in the rainy season. In contrast, this paper quantitatively examines the wealth gradient in drinking water quality in the rainy season and finds a differential wealth gradient in the pooled data. However, I find that there is heterogeneity at the country level where some countries experience the wealth gradient getting steeper and others experience the wealth gradient getting flatter. In Benin, Central African Republic, and DR Congo the rainy season makes the wealth gradient steeper. For these countries, among those households surveyed in the rainy season, a household being in the richest quintile is associated with a 6.3 to 24.8 percentage point increase in the probability of having improved drinking water compared to the poorest

² Kostyla (2015) identified 22 studies in developing countries through a systematic review and analyzed seasonal variation in fecal contamination.

households. However, in Guinea, Lesotho, and Madagascar the wealth gradient changes in a different way. For these countries, among households surveyed in the rainy season, being in the richest quintile is associated with a 17.8 to 27.1 percentage point decrease in the probability of having improved drinking water compared to the poorest households.

To address the issue of water quality crisis, studies have evaluated the effectiveness of a wide variety of interventions across a variety of contexts.³ Systematic reviews suggest that households do one of two things to attempt to improve water quality at point of use (Clasen 2007). According to the review, some households use better quality sources to have better water at point of use. Whereas others decide to treat their water prior to consumption (Kremer 2021). I observe both the decision to treat water prior to consumption and the selection of water source in my dataset. To study why there is a wealth gradient in drinking water quality, I explore a household's choice of water source and the decision to treat their water as potential mechanisms.

I find that being in the richer wealth quintiles is associated with an increase in the probability of water treatment on average in the dry season. In the dry season, being in the fourth and fifth wealth quintile is associated with a 3.0 and 3.8 percentage point increase in the probability of water treatment when compared to households in the poorest wealth quintile. However, the rainy season does not appear to be associated with a household's water treatment decision in the pooled data.

Households also endogenously choose their source of water to have better quality drinking water (Clasen 2007). In the pooled data, richer households on average use better quality water sources and in the rainy season, decrease their usage of surface water sources to shift to using basic water sources even more. A household being in the richest quintile is also associated

³ Kremer (2021) conducted a meta-analysis of 15 RCTs on water treatment and Clasen (2007) conducted a meta-analysis 32 studies on interventions to improve water quality.

with a 6.1 percentage point drop in the probability of using surface water compared to households in the poorest wealth quintile in the dry season. A household being in the richest wealth quintile is also associated with a 13.1 percentage point increase in the probability of using basic water sources. Among those surveyed in the rainy season, being in the second, third and fourth quintile is associated with a 4.2, 6.5, and 7.0 percentage point decrease in surface water usage respectively compared to households in the poorest quintile.

These patterns suggest that there is a wealth gradient in drinking water quality in African countries. The endogenous decision to treat water prior to consumption and selection of water sources are mechanisms that contribute to the wealth gradient. I find no differential wealth gradient in my pooled data. However, I find that there is a differential wealth gradient at the country level; but because there is heterogeneity in how the wealth gradient changes at the country level, the effect is close to zero in the pooled data. My paper compliments, and adds to, the literature on the determinants of water quality in households. Jessoe (2013) for example, focuses on the substitutability between home treatment and choice of water source. I also contribute to the literature on unequal allocation of drinking water quality across wealth quintiles. In 2019, WHO published a report stating wealthier households and areas often getting the benefits of subsidies to improve access to water potentially creating a wealth gradient (WHO 2019). In contrast, my contribution is to show that there is a wealth gradient in drinking water quality in Africa and that there is a differential wealth gradient in the rainy season. I show that there is heterogeneity in the wealth gradient at the country level. Finally, I contribute to the literature on unintended degradation of water quality from the source to the point of use. To the best of my knowledge, my paper is the first to use the difference in risk levels between the source and point of use as a measure of water quality.

It is important to note that this paper has potential limitations. The findings in this paper are subject to unobservable variables at both the household and at the country level that might be confounded with drinking water quality. Additionally, the findings in this study might not generalize to countries that are in different stages of development and growth compared to countries in my African sample. I also use just one dimension of having clean drinking water: the presence of e. coli. The results may not be robust to other definitions of having clean water.

The remainder of the paper proceeds as follows: Chapter two presents the background and describes the data for the sample. Chapter three presents the evidence on the wealth gradient in the African household water quality in dry and rainy seasons, and Chapter four tests decisions of water treatment and water source as mechanisms to explain the wealth gradient in drinking water quality. Chapter five concludes.

CHAPTER TWO

Background and Data Description

This section first provides background information on clean drinking water in Africa, E. coli contamination, and the effects of the rainy season on access to clean drinking water. characteristics and water quality. Second, I provide information on the wealth index being used, the water sources, and my outcome variables of interest. Third, I provide information on the data I use and descriptive statistics of household level.

2.1 Access to Clean Drinking Water in Africa

Access to clean drinking water remains a critical challenge across many African countries, directly impacting public health, economic development, and social well-being. The number of people without access to water is growing in Sub-Saharan Africa. It is the only region of the world where this is happening. About 387 million people lived without access to basic drinking water services in Sub-Saharan Africa in 2020, up from 350 million people in 2000 (WHO/UNICEF 2021). Despite significant global progress towards improving water access, disparities persist within and between African nations, driven by factors such as geographical location, infrastructure, socio-economic status, and governance. This backdrop is essential for understanding the nuances of water quality variation across different wealth quintiles within these countries.

WHO guidelines recommend that drinking water should be free from pathogens and elevated levels of harmful substances at all times (WHO 2017). Globally, the highest priority parameters for health are fecal contamination, arsenic, and fluoride. The presence of *Escherichia*

coli (*E. coli*) is considered a reliable indicator of fecal contamination. I use the presence of *E. coli* per 100 ml water to derive my measurements of water quality in this paper.

2.2 Water Quality in the Rainy Season

Seasonal variations, particularly the rainy season, exacerbate existing disparities in water quality. Rainfall can lead to increased runoff and contamination of water sources, disproportionately affecting those who rely on surface water or unprotected wells. Kostyla et al. (2015) found that fecal contamination is more often detected and found at higher concentrations during the wet season. This is consistent with the understanding that microbial contamination varies with time (Edberg et al., 2000) and confirms the widely accepted perception that fecal contamination is generally greater in the wet season (WHO/UNICEF, 2010) to be a real and significant pattern of variation. The greater fecal contamination is why I also analyze the wealth gradient in having clean water in the rainy season as well as the dry season to understand the seasonal variation in the wealth gradient.

2.3 Data

To investigate the wealth gradient in drinking water quality at point of consumption, I use the Multiple Indicator Cluster Survey (MICS) datasets from the sixth round (MICS6), encompassing 14 African countries that have the rapid water quality testing module. UNICEF implemented these surveys in collaboration with national ministries of health and statistics offices and conducted them between 2017 and 2021. The MICS surveys are cross-sectional and they were collected with a stratified two-stage sampling design in order to be nationally representative.

The MICS6 datasets are the first MICS datasets to include the rapid water-quality testing module. The households in the survey were randomly selected to be tested for water quality. The households selected for water quality testing had drinking water samples from the household and where the household collected the water both tested. Both these water samples were analyzed for the presence of *Escherichia coli* (*E. coli*), an indicator of fecal contamination.

2.4 Water Quality Module and Quality Assessment in the MICS Surveys

The MICS is a survey program supported by UNICEF. The drinking water quality module for the survey was developed as a collaboration between the JMP and the global MICS program. The module was conducted in a subsample of households (between 3 and 5) within each cluster.

In households selected for water quality testing, the interviewer requested permission to collect samples of drinking water, asking the respondent for a “glass of drinking water” (i.e., the PoU) and to be shown the location of the source of the drinking water (i.e., the PoC) to take a sample from that location. The two samples were analyzed to examine differences in quality between the PoC and the PoU (Shields et al. 2015; Wright et al. 2004).⁴

Water samples are analyzed for *E. coli* using a portable membrane filtration apparatus. *E. coli* is measured by field teams, who filter 100mL of sample through a μ 0.45- μ m filter (Millipore Microfil), which is then placed onto CompactDry EC growth media plates (Nissui) that are rehydrated with 1mL of sample water. For the MICS6 surveys, “*E. coli* only” CompactDry plates are used, with the advantage that a second media plate is not needed to count up to 100 *E. coli*.

⁴ An international water quality trainer working alongside national experts led the training of the field teams. National experts from regulatory agencies, research laboratories, or ministries of health provided technical input to the training and oversight during the fieldwork through a small number of field visits to allow for observation of the testing procedure.

The levels of *E. coli* are classified according to the number of colony forming units (CFUs) per 100mL as follows: <1, low risk; 1–10, moderate risk; 11–100, high risk; and >100, very high risk (WHO/UNICEF 2012; WHO 2017).

2.5 Construction of the Wealth Quintile Variables

In the Multiple Indicator Cluster Surveys (MICS), the wealth quintile variable is constructed using a composite measure of a household's cumulative living standard. A principal component analysis (PCA) is carried out which assesses the ownership of various assets, such as televisions and bicycles, access to utilities like water and electricity, and the quality of the dwelling's construction materials. The PCA generates a wealth score for each household, reflecting its overall economic status relative to the broader population within the survey.

Following the generation of wealth scores, households are then ranked from the poorest to the wealthiest. The ranked list is divided into five equal parts, each representing 20% of the sampled population, thus forming the wealth quintiles. These quintiles are labeled from the first (poorest) to the fifth (richest), allowing for comparative analysis across different levels of economic well-being. This categorization is important in assessing disparities in access to resources, health outcomes, and, crucially for this paper, variations in water quality and access across different socio-economic groups within African countries.

2.6 Drinking water Sources

Drinking water services refers to the accessibility, availability and quality of the main source used by households for drinking, cooking, personal hygiene, and other domestic uses (JMP 2022). Improved drinking water sources are those which, by nature of their design and construction, have the potential to deliver safe water. The population using improved sources can be subdivided into three groups according to the level of service provided. In order to meet the

criteria for a safely managed drinking water service, households must use an improved source that is: accessible on premises, available when needed, and free from contamination. If the improved source does not meet any one of these criteria but a round trip to collect water takes 30 minutes or less, then it is classified as a basic drinking water service. If water collection from an improved source exceeds 30 minutes it is categorized as a limited service. Populations using unimproved sources such as unprotected wells or springs, and populations drinking surface water collected directly from a river, dam, lake, stream, or irrigation canal are also subdivided.

2.7 Outcome Variables

For this analysis, the primary variable of interest is a measure of having clean drinking water. I define that in the following way: I construct a binary variable, “Improved Water” that takes the value of 1 if a household has better water at the point of consumption than point of use. The water is classified at 3 risk levels: Low risk, Moderate to high risk and very high risk. If the risk level of the water at point of use is lower than the risk level of water at the source, then I consider a household to have improved water.⁵ The risk levels are based on E. coli levels in the water and are based on colony forming units (CFUs) per 100mL. They are defined as follows: <1, low risk; 1–100, Moderate to high risk; and >100, very high risk (WHO/UNICEF 2012; WHO 2017).

2.8 Sample and Descriptive Statistics

My pooled sample comprises of 27,408 households across 14 African countries with water quality data.⁶ Household level summary statistics for my key variables are presented in

⁵ Appendix A Table 1 shows how the water quality variable is constructed based on the risk level at source and point of use.

⁶ Appendix A Table 2 lists the countries and presents country level summary statistics.

Table 1 and show an average size of 4.87 members per household. On average, there are 0.75 children under five years old in these households. A significant proportion, 32 percent, reside in urban settings, with the majority, 71 percent, having a literate head of household. Handwashing facilities with soap and water are present in 42 percent of households, while 28 percent primarily practice open defecation. In contrast, a mere 11 percent have access to flush toilets. The task of water collection is predominantly undertaken by females, as only 20 percent of households report a male collector. On average, the collector is 30.05 years old. Water collection is a routine yet time-consuming task, with an average round trip taking 24.78 minutes and occurring roughly 11.79 times over the past week. Only 10 percent of households treat their water before consumption. E. coli levels average 35 units at the source and 50.26 units at point of consumption, indicating ineptitude in water quality management.

On average, 55.7 percent of the households in my sample have basic water service. Among the households surveyed, 16.4 percent rely on limited water service, 19.5 percent of the households rely on unimproved water service, and 7.3 percent of the people rely on surface water on average. Table 1, Column 3 shows that in the rainy season, the percentage of households using basic, limited, and surface water goes up. However, the percentage of households using unimproved water sources goes down. Among households surveyed in the rainy season, 53.6 percent have basic water service, 15.6 percent have limited water service, 23.4 percent have unimproved water service and 6.3 percent have surface water service. Note that the percentage of households using basic and limited water service goes down and the percentage of households using unimproved water service goes up. On average, more households are relying on lower quality water among those households surveyed in the rainy season.

CHAPTER THREE

Relationship Between Wealth Quintile and Improved Water

The goal of this paper is to study the disparity in water quality by wealth quintile. Figure 3.1 presents the overall wealth gradient in drinking water in the pooled data. To study the wealth gradient, I first look at the correlates of improved water at point of use and the differential effects of the rainy season on having improved water at point of use separated by each wealth quintile.

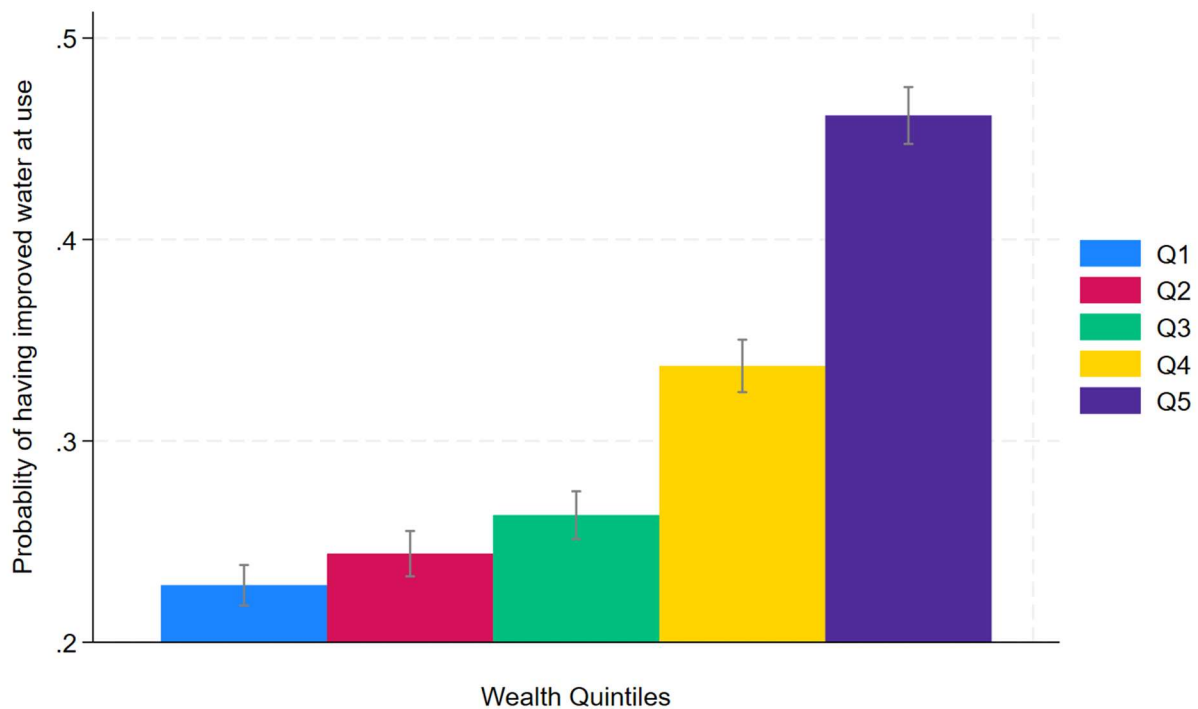


Figure 3.1. The figure depicts the probability of having improved water at point of use for 14 African countries by the wealth index.

3.1 Correlates of improved Water at Point of Use

Methods

To study the correlates of having better water at point of use, I use a multivariate OLS regression method. I run the following regression:

$$(1) \text{WaterQuality}_{ic} = \alpha + \beta X_{ic} + \varepsilon_{ic}$$

where WaterQuality_{ic} represents the outcome variables of interest for household i in country c : an indicator variable constructed to represent whether the household has a better category water at point of use at use than the source, water quality at point of use measured by levels of e. coli and water quality at source also measured by level of e. coli. The error term ε_{ic} captures unobservable factors that may affect the quality of the water.

X_i is a vector that consists of various household characteristics that might impact the quality of the water at the point of use. The variables of interest are categorical variables for the source a household gets their water from (Basic, Limited, Unimproved, and surface water)⁷ where surface water is the omitted category, categorical variables for the type of sanitation services a house has (Basic, limited, unimproved, and open defecation) where open defecation is the omitted category⁸, the characteristics of the water carrier (sex, age), categorical variables for wealth quintile (poorest, second poorest, middle and fourth) where richest is the omitted category, an

⁷ Basic services include piped water, boreholes or tubewells, protected dug wells, protected springs, rainwater, and packaged or delivered water. Limited refers to an improved source more than 30 minutes roundtrip. Unimproved sources include unprotected dug wells and unprotected springs. Surface Water (No service) refers to the direct collection of water from surface waters such as rivers, lakes or irrigation channels.

⁸ At least basic (Basic Service) sanitation services (SDG 1.4.1) refer to the use of improved facilities which are not shared with other households. Improved sanitation facilities are those designed to hygienically separate excreta from human contact and include flush/pour flush to piped sewer system, septic tanks or pit latrines; ventilated improved pit latrines, composting toilets or pit latrines with slabs. Limited sanitation service refers to an improved facility shared with other households. Unimproved sanitation facilities include flush/pour flush to an open drain, pit latrines without a slab, hanging latrines and bucket latrines. Open defecation refers to No service.

indicator variable for if the household head is literate, if the household has soap and water, if the household is in an urban area, and the size of the household (total and number of children).

Results

Table 2 presents the results of the correlates of having improved water at point of use based on Equation 1. Column 1 shows the correlates of having better water, Column 3 shows the correlates of water quality at use and Column 5 shows the correlates of water quality at source. Column 2, 4, and 6 add controls and country fixed effects to the regression so that the estimates compared are making within country comparisons. The table shows that the choice of water source for a household has no association with having improved water at point of use. However, having basic and limited water services as the source of water is associated with decreasing the units of bacteria in the source water. A household being in a lower quintile is associated with decreasing the probability of having improved water at point of use. Additionally, being in a lower wealth quintile is associated with higher units of bacteria both at point of use and source.

Table 2, Column 2 shows that being in the poorest wealth quintile is associated with a 22.1 percentage point decrease in the probability of having improved water at point of use compared to the richest households. Being in the second quintile is associated with a 19.0 percentage point decrease in the probability of having better quality water compared to the richest quintile. Being in the middle quintile is associated with a 17.3 percentage point decrease in the probability of having improved water and being in the fourth quintile is associated with a 12.3 percentage point decrease in the probability of having improved water compared to households in the richest quintile. All the coefficients are significant at the 1 percent level.

Table 2, Column 4 presents the correlates of water quality at point of use and Column 6 presents the correlates of water quality at source. These columns show that being in the poorer

wealth quintiles is associated with higher units of bacteria presence both at point of use and at source when compared to the richest households. Being in the poorest quintile is associated with a 21.3 unit increase in the number of bacteria at use and a 10.0 unit increase in in the number of bacteria at source when compared to households in the richest quintile.

3.2 Wealth Gradient in Having Improved Water During the Rainy Season

Figure 3.2 illustrates the differential wealth gradient in rainy and non-rainy seasons. The figure plots the probability of having improved water on the y-axis and each wealth quintile on x-axis separated by the season they were surveyed in. The figure presents that the rainy season on average lowers the probability of having better quality water considerably for the poorest households compared to the richest households in the pooled data without country fixed effects.

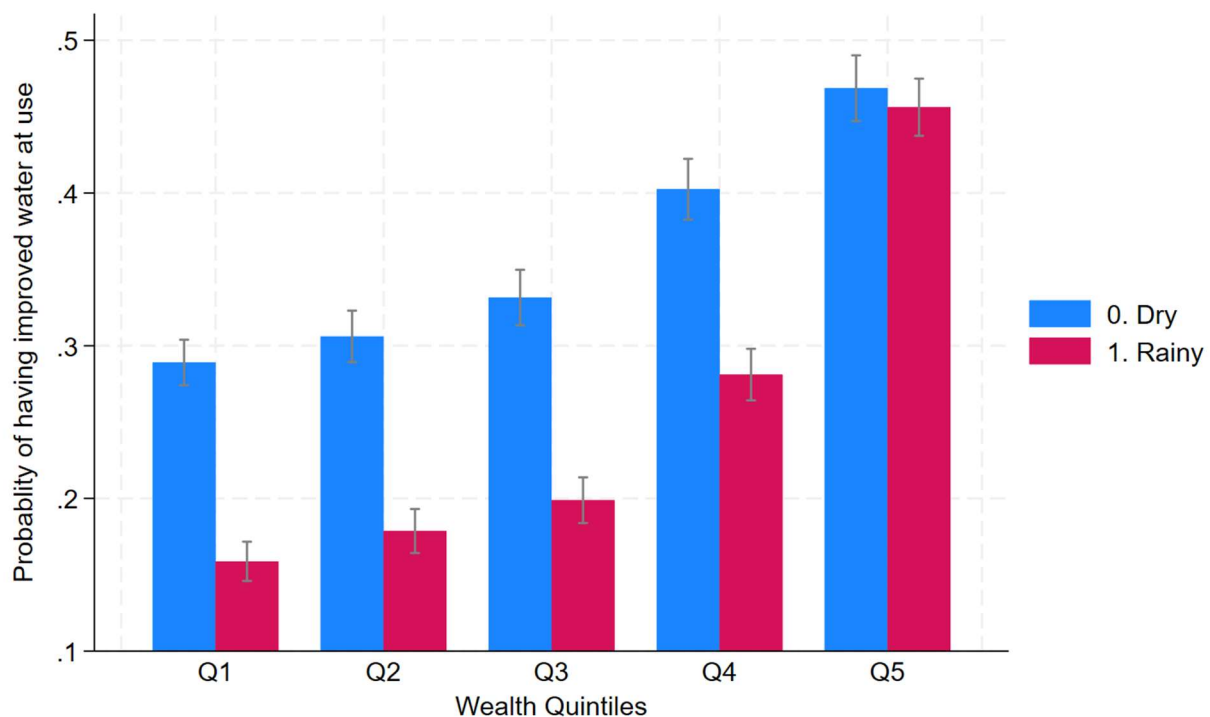


Figure 3.2. The figure depicts the probability of having improved water at point of use for 14 African countries by the wealth index separated by the timing of the survey. Rainy indicates that a household was surveyed during the rainy season of a country and dry indicates otherwise.

Methods

To study how being surveyed in the rainy season might be correlated with water quality, I run the following regression:

$$(1) \text{WaterQuality}_{ic} = \alpha + \beta_1 \text{Rainy}_{ic} + \beta_2 Q2_{ic} + \beta_3 Q3_{ic} + \beta_5 Q4_{ic} + \beta_6 Q5_{ic} + \beta_7 \text{Rainy} * Q2_{ic} + \beta_8 \text{Rainy} * Q3_{ic} + \beta_9 \text{Rainy} * Q4_{ic} + \beta_{10} \text{Rainy} * Q5_{ic} + \beta_i X_{ic} + \varepsilon_{ic}$$

where WaterQuality_{ic} are my outcome variables: an indicator variable that takes the value of 1 if a household has improved water at point of use than at source, the level of bacteria in the water at point of use, and the level of bacteria in the source water. The key regressors are the interaction of the rainy season dummy with an indicator variable for each of the wealth quintiles where poorest is the omitted category. The coefficients on these interaction terms capture the differential outcome for each wealth quintile during the rainy season. I estimate this regression with and without controls and country fixed effects.

Results

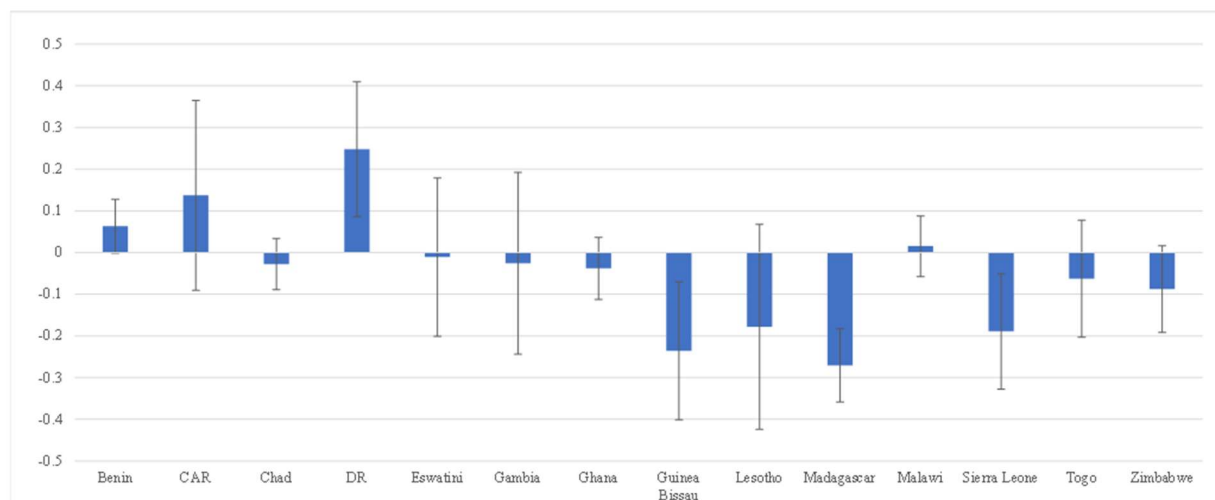
Table 3 reports the results. There are two main points of the table: first, the coefficients on the wealth quintile indicators show a clear wealth gradient in the dry season. Second, the coefficients on the interaction terms show that there is no differential wealth gradient in the rainy season in the pooled data. The reason the coefficients on the interaction term show no differential wealth gradient in the rainy season is because there is heterogeneity at the country level. In some countries, the wealth gradient gets steeper and in others, it gets flatter.⁹

Table 3, Column 1 reports the correlates of having improved water in the rainy season with household level controls. The coefficient on fifth * rainy shows that during the rainy season, a

⁹ Appendix A Tables 3 through 16 present the coefficients for each country.

household being in the richest quintile is associated with a 11.8 percentage point increase in the probability of having improved water at point of use compared to households in the poorest quintile. However, Column 2 shows that there is no differential wealth gradient when I add country fixed effects to the regression. This happens due to heterogeneity at the country level. Appendix B tables 3 to 16 show that the wealth gradient changes in different ways in the rainy season based on the country a household is in.

In Table 3, Column 2, the coefficient on third * rainy and fourth * rainy show that among households surveyed in the rainy season, being in the third and fourth quintile is associated with a 4.2 and 5.3 percentage point decrease in the probability of having improved water compared to the poorest households. Both of these coefficients are significant at the 1 percent level. However, the coefficient on fifth * rainy and second * rainy is no longer significant indicating that there is no differential wealth gradient in the rainy season in the pooled dataset. This happens due to the rainy season having heterogeneous effects at the country level.



*Notes: The figure plots the coefficient of fifth * Rainy at the country level running the regression from equation 2 separately for each country.*

Figure 3.3: Differential Wealth Gradient in the Rainy Season at the Country Level

Figure 3.3 presents the heterogeneity at the country level during the rainy season. As the figure shows, in some countries the wealth gradient becomes steeper and in others the wealth gradient becomes flatter. In Benin, Central African Republic, and DR Congo the rainy season makes the wealth gradient steeper. For these countries, among those households surveyed in the rainy season, a household being in the richest quintile is associated with a 6.3 to 24.8 percentage point increase in the probability of having improved drinking water. However, in Guinea, Lesotho, and Madagascar the wealth gradient changes in a different way. For these countries, among households surveyed in the rainy season, being in the richest quintile is associated with a 17.8 to 27.1 percentage point decrease in the probability of having improved drinking water.

CHAPTER FOUR

Hypotheses: Testing Different Mechanisms – Water Source or Water Treatment?

The analysis above finds that there is a wealth gradient in drinking water quality in African countries. Households can make a number of behavioral choices that impact water quality. Jessoe (2013) finds that a household's behavioral choices of water treatment and water source effect water quality. In the data, I observe households improving their water in both ways. Can my data provide an insight as to why a household being in the richest wealth quintile is associated with a higher likelihood of having improved drinking water? I explore this by looking at two things. First, I analyze the wealth gradient in water treatment in the dry and rainy season and then I analyze the wealth gradient in water source selection in the dry and rainy season.

4.1 Wealth Gradient in Water Treatment

One reason the rainy season might have differential effects by wealth quintiles might be because of variation in treatment practices. Water treatment is a critical intervention for improving the quality of water at the point of use, particularly in contexts where source water quality is compromised (Kremer 2021). However, the adoption and effectiveness of water treatment practices can vary significantly across different wealth quintiles. I hypothesize that poorer households are less likely to engage in effective water treatment practices in the rainy season, leading to higher levels of contamination and lower probability of having improved water compared to wealthier households.

To test this prediction, I rerun Equation 2 with water treatment as my outcome variable. The variable is a binary indicator variable that takes the value of 1 if a household treats their

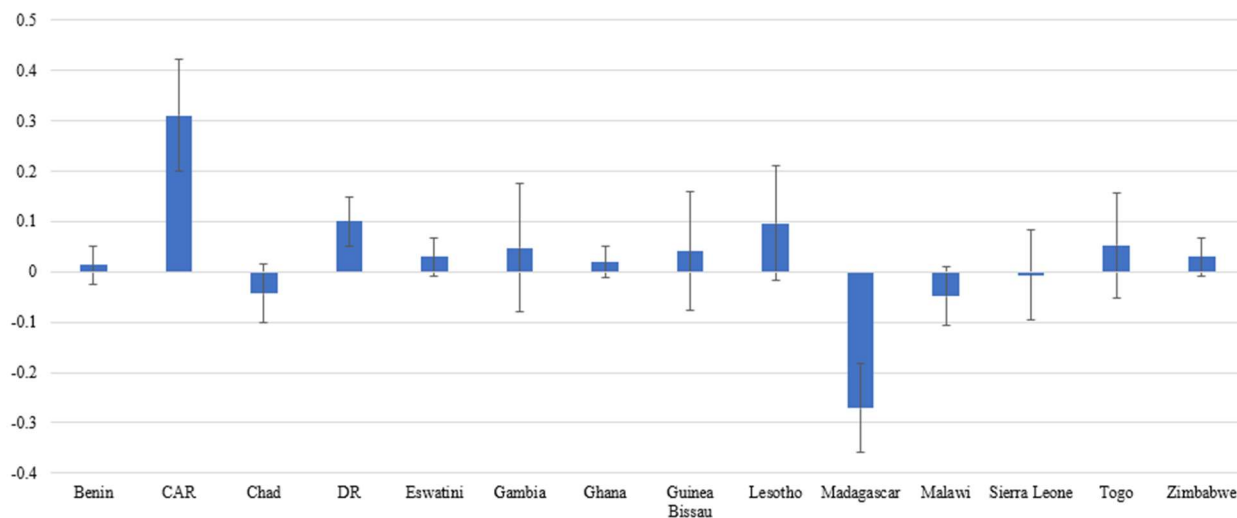
water prior to consumption and 0 otherwise. I run the regressions with just the interaction terms, the interactions terms with controls, and the interaction terms with controls and country fixed effects. If the coefficients on the richer wealth quintiles is higher in magnitude than the coefficients on the lower wealth quintiles, then that means that there is a wealth gradient in water treatment.

Table 4 presents the results. Column 1 presents the results of the regression with just the interaction terms, Column 2 with household level controls and Column 3 with both controls and country fixed effects. The table shows that being surveyed in the rainy season does not differentially impact the water treatment decision of households in different wealth quintiles. Therefore, this does not explain the steeper wealth gradient in the rainy season.

When making comparisons within the same country, being in the second or the third quintile has no discernible impact on water treatment decisions. However, being in the fourth and fifth quintile is positively associated with water treatment. Table 4, Column 3 shows that a household being in the fourth quintile is associated with a 3-percentage point increase in water treatment and a household being in the fifth quintile is associated with a 3.8 percent increase in water treatment compared to households in the poorest quintile. Therefore, when I compare households within the same country, richer households are more likely to treat their water than poorer households.

In Table 4, Column 3, the coefficient on richest * rainy shows that during the rainy season, a household being in the richest quintile is associated with a 0.1 percentage point increase in the probability of treating the water prior to consumption compared to the poorest households. Similarly, the coefficient on second * rainy shows that during the rainy season, a household being in the second quintile is associated with a 0.5 percentage point decrease in the

probability of having basic water services. When I run the regression with country fixed effects, the rainy season does not appear to have any discernible impact on a household's decision to treat their water across the wealth quintiles. There appears to be no effects when I run the regression with country fixed effects because of heterogeneity at the country level as seen in Figure 4.1.



*Notes: The figure plots the coefficient of fifth*Rainy at the country level running the regression from equation 2 separately for each country.*

Figure 4.1: Differential Wealth Gradient in the Rainy Season at the Country Level

Figure 4.1 shows heterogeneity in the wealth gradient seen in the rainy season on water treatment behavior at the country level. The wealth gradient gets steeper in countries like Central African Republic and Gambia and gets flatter in Madagascar. This means that water treatment decisions play a role in the wealth gradient in drinking water quality in African countries.

4.2 Wealth Gradient in Water Sources

Another mechanism that might explain why households in the richer quintiles are more likely to have improved water might be their choice of water sources. The utilization of improved water sources—such as piped water, protected wells, and rainwater harvesting

systems—significantly enhances the quality of water at the point of use compared to more traditional or unprotected sources (Classen 2007). Improved sources are designed to reduce direct exposure to environmental contaminants and pathogens, thereby inherently providing safer water. These sources often incorporate physical barriers to contamination, such as sealed piping or protective coverings, and may include initial treatment processes that further ensure water safety before it reaches the consumer.

The key mechanism through which improved water sources enhance water quality lies in their ability to minimize the entry of pollutants and microorganisms at the point of collection. This is in contrast to unprotected sources, such as open wells and surface water, where water is readily exposed to environmental contaminants, animal waste, and other sources of pollution. As a result, water from improved sources typically requires less intensive treatment at the household level to be considered safe for consumption.

The hypothesis, therefore, is that households in the richer wealth quintiles on average, use better water sources. That would explain why they have a higher likelihood of having improved water at point of use. To test this prediction, I rerun Equation 2 with water sources as my outcome variables. The variables are binary indicator variables constructed to represent what kind of water source a household uses to acquire drinking water from. The outcome variables of interest are: Basic water service, Limited water service, Unimproved water service and surface water. I run the regressions with just the interaction terms and the interaction terms with controls and country fixed effects. If the coefficients on the interaction terms with the richer wealth quintiles is higher in magnitude than the coefficients on the interaction terms with poorer wealth quintiles for better water sources, then that means that richer households use better water sources

than poorer households in the rainy season. That would confirm my hypothesis that water source is a mechanism for the steeper wealth gradient.

Table 5 reports the results. The coefficients on fourth and fifth quintile show that a household being in the fourth and fifth quintile is associated with a 12.9 and 13.1 percentage point increase in the probability of using a basic water source respectively compared to the poorest households. On the other hand, a household being in the fourth and fifth quintile is associated with a 5.6 and 6.1 percentage point decrease in the probability of using a surface water source respectively compared to the poorest households. This shows that one mechanism for the existing wealth gradient is households in wealthier quintiles being able to rely more on better water sources.

Table 5, Column 2 shows that among households surveyed in the rainy season, being in the richest quintile is associated with a 4.8 percentage point increase in the probability of using basic water services and a 6.8 percentage point decrease in the probability of using surface water compared to the poorest households. Table 5, Column 8 shows that among households surveyed in the rainy season, being in the second, middle and fourth quintile is associated with a 4.2, 6.5, and 7.0 percentage point decrease in the probability of using surface water respectively compared to the poorest households. However, a household being in those quintiles is not associated with an increase in the probability of using basic water service unlike the richest households.

These results show that households on average reduce their usage of surface water in the rainy season. However, the richest households shift to using basic water services whereas households in the other four quintiles do not make the switch. Thus, the decision of water source selection is a mechanism that contributes to the wealth gradient in these African countries.

Additionally, the decision of water source selection might explain the heterogenous wealth gradient in the rainy season by country.

CHAPTER FIVE

Conclusion

This paper examines the wealth gradient in water quality in dry and rainy seasons in 14 countries. I document the following three main facts and am the first to do so. First, there is an important wealth gradient in the likelihood of having improved drinking water at point of use. Second, the rainy season reduces drinking water quality across all sources and it differentially affects the wealth gradient. However, there is variation across countries in how the wealth gradient changes in the rainy season. Third, I explore the reasons why the wealth gradient exists and find that richer households are more likely to have improved water sources and are more likely to treat their water.

A household being in a higher wealth quintile is positively associated with the probability of having improved drinking water. First, I find that a household being in the richest wealth quintile is associated with a 21.2 percentage point increase in the probability of having improved drinking water compared to the poorest households in the dry season. A household being in the fourth wealth quintile is associated with a 12.9 percentage point increase in the probability of having an improved water source when compared to households in the poorest quintile in the dry season.

I find that the rainy season makes the overall quality of the water worse across all the countries in my sample. Being surveyed in the rainy season is associated with a 6.7 unit increase in the amounts of bacteria in water at point of use and 6.3 unit increase in the number of bacteria at source. However, being surveyed in the rainy season does not have any association with the probability of having improved drinking water for any household. Among households

surveyed in the rainy season, being in the third and fourth quintile is associated with a 4.2 and 5.3 percentage point decrease in the probability of having improved water respectively. Among households surveyed in the rainy season, being in the richest quintile is associated with a 2.2 percentage point increase in the probability of having improved water. However, none of these coefficients are statistically significant. This is evidence that there is no differential wealth gradient in the rainy season in the pooled dataset. However, there is heterogeneity in the differential wealth gradient across countries. In Benin, Central African Republic, and DR Congo the rainy season makes the wealth gradient steeper. For these countries, among those households surveyed in the rainy season, a household being in the richest quintile is associated with a 6.3 to 24.8 percentage point increase in the probability of having improved drinking water. However, in Guinea, Lesotho, and Madagascar the wealth gradient changes in a different way. For these countries, among households surveyed in the rainy season, being in the richest quintile is associated with a 17.8 to 27.1 percentage point decrease in the probability of having improved drinking water.

Then I present two facts that might explain this wealth gradient in drinking water quality. First, I test the hypothesis that richer households are more likely to treat their water before consumption. I find that being in the fourth and fifth wealth quintile is associated with a 3.0 and 3.8 percentage point increase in the probability of water treatment when compared to the households in the poorest wealth quintile. Therefore, I find that water treatment is a mechanism behind the wealth gradient.

Second, richer households on average are more likely to use improved water sources compared to households in the poorer wealth quintiles. A household being in the richest quintile is also associated with a 6.1 percentage point drop in the probability of using surface water

compared to households in the poorest wealth quintile. A household being in the richest wealth quintile is also associated with a 13.1 percentage point increase in the probability of using basic water sources compared to the poorest households. This pattern of having better water sources in combination with the endogenous decision to treat their water more allows households in the richest wealth quintiles to have a higher likelihood of improved drinking water.

It is important to note that this paper has potential limitations. The findings in this paper are subject to unobservable variables at both the household and at the country level that might be confounded with drinking water quality. Additionally, the findings in this study might not generalize to countries that are in different stages of development and growth compared to countries in my African sample. It is important to understand the differential wealth gradient in the rainy season across countries that I do not address in this paper and try to understand the underlying mechanisms.

Table 1- Summary Statistics and Balancing

	All	Dry season	Rainy season	Difference	
	(1)	(2)	(3)	(4)	
Improved water	0.309 (0.003)	0.357 (0.004)	0.264 (0.004)	0.093 (0.005)	***
Source water test (100ml)	32.935 (0.249)	28.514 (0.341)	37.116 (0.358)	-8.603 (0.495)	***
Basic water service	0.557 (0.003)	0.536 (0.004)	0.577 (0.004)	-0.040 (0.006)	***
Limited water service	0.164 (0.002)	0.156 (0.003)	0.172 (0.003)	-0.016 (0.004)	***
Unimproved water service	0.195 (0.002)	0.234 (0.004)	0.158 (0.003)	0.075 (0.005)	***
Surface water service	0.073 (0.002)	0.063 (0.002)	0.081 (0.002)	-0.018 (0.003)	***
Male water carrier	0.195 (0.002)	0.197 (0.003)	0.194 (0.003)	0.003 (0.004)	
Age of water carrier	30.051 (0.077)	30.705 (0.113)	29.432 (0.104)	1.273 (0.153)	***
First quintile	0.242 (0.003)	0.262 (0.004)	0.222 (0.003)	0.040 (0.005)	***
Second quintile	0.203 (0.002)	0.212 (0.003)	0.195 (0.003)	0.017 (0.005)	***
Third quintile	0.193 (0.002)	0.193 (0.003)	0.194 (0.003)	-0.001 (0.005)	
Fourth quintile	0.185 (0.002)	0.175 (0.003)	0.194 (0.003)	-0.019 (0.005)	***
Fifth quintile	0.177 (0.002)	0.158 (0.003)	0.195 (0.003)	-0.037 (0.005)	***
HH head literate	0.712 (0.003)	0.725 (0.004)	0.698 (0.004)	0.027 (0.005)	***
Flush	0.120 (0.002)	0.090 (0.002)	0.148 (0.003)	-0.058 (0.004)	***
Open defecation	0.275 (0.003)	0.278 (0.004)	0.273 (0.004)	0.004 (0.005)	
Soap and water in HH	0.422 (0.002)	0.407 (0.003)	0.435 (0.003)	-0.028 (0.004)	***
HH in urban area	0.330 (0.003)	0.334 (0.004)	0.326 (0.004)	0.008 (0.006)	
Number of HH members	4.868 (0.018)	5.002 (0.027)	4.741 (0.023)	0.262 (0.035)	***
Number of children under age	0.752 (0.006)	0.764 (0.008)	0.740 (0.007)	0.025 (0.011)	**
	28420	13812	14608	28420	

Notes: The means for the specified variables are calculated at the household level. Rainy and Dry season represent the timing of the survey. Standard deviation in parentheses. Column 4 presents the difference between the dry season and the rainy season.

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Table 2 - Correlates of Having Improved Water at Use

	Improved water		Water quality at use		Water quality at source	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Wealth index (Richest omitted)</i>						
First (Poorest)	-0.137*** (0.010)	-0.227*** (0.010)	12.296*** (0.948)	25.481*** (0.846)	12.548*** (0.907)	19.252*** (0.854)
Second	-0.130*** (0.010)	-0.196*** (0.010)	12.697*** (0.953)	22.310*** (0.845)	11.628*** (0.912)	15.890*** (0.853)
Third (Middles)	-0.130*** (0.010)	-0.178*** (0.009)	11.322*** (0.920)	18.320*** (0.811)	9.589*** (0.881)	12.354*** (0.819)
Fourth	-0.093*** (0.009)	-0.125*** (0.009)	6.960*** (0.870)	11.443*** (0.763)	5.945*** (0.833)	8.006*** (0.771)
HH head literate	0.100*** (0.006)	-0.007 (0.007)	-19.448*** (0.592)	-2.743*** (0.583)	-19.661*** (0.567)	-3.425*** (0.589)
Urban	0.086*** (0.007)	0.068*** (0.007)	-11.259*** (0.656)	-8.441*** (0.596)	-7.438*** (0.628)	-9.136*** (0.602)
Number of people in HH	-0.007*** (0.001)	-0.009*** (0.001)	0.820*** (0.106)	0.889*** (0.094)	0.633*** (0.102)	0.410*** (0.095)
Number of children in HH	-0.013*** (0.004)	-0.003 (0.003)	2.149*** (0.338)	1.054*** (0.300)	2.395*** (0.324)	1.141*** (0.303)
Constant	0.346*** (0.012)	0.313*** (0.013)	52.222*** (1.093)	54.073*** (1.133)	36.112*** (1.047)	42.979*** (1.145)
Observations	28,420	28,420	28,420	28,420	28,420	28,420
HH Level Control	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed effects	No	Yes	No	Yes	No	Yes
R-squared	0.054	0.166	0.107	0.319	0.100	0.237
Mean dep var	0.31	0.31	49.34	49.34	32.93	32.93

Notes: Standard error in parentheses. Water quality is measured by the levels of E. coli or number of colony forming units (CFUs) per 100mL. Basic services include: piped water, boreholes or tubewells, protected dug wells, protected springs, rainwater, and packaged or delivered water. Limited refers to an improved source more than 30 minutes roundtrip. Unimproved sources include unprotected dug wells and unprotected springs. Surface Water (No service) refers to the direct collection of water from surface waters such as rivers, lakes or irrigation channels.

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Table 3 - Wealth Gradient in Water Quality in the Rainy Season

	Improved water		Water quality at use		Water quality at source	
	(1)	(2)	(3)	(4)	(5)	(6)
Second wealth quintile	0.011 (0.011)	0.040*** (0.011)	1.055 (1.042)	-2.711** (0.930)	0.101 (1.012)	-2.246* (0.939)
Third wealth quintile (Middle)	0.018 (0.012)	0.070*** (0.011)	-0.118 (1.084)	-6.991*** (0.971)	-2.368* (1.054)	-6.386*** (0.981)
Fourth wealth quintile	0.057*** (0.012)	0.129*** (0.012)	-3.963*** (1.151)	-14.655*** (1.037)	-3.830*** (1.118)	-10.139*** (1.048)
Fifth wealth quintile (Richest)	0.096*** (0.013)	0.212*** (0.013)	-7.051*** (1.249)	-23.896*** (1.142)	-10.179*** (1.213)	-19.317*** (1.154)
Rainy season	-0.120*** (0.011)	0.016 (0.011)	22.419*** (1.018)	6.703*** (0.997)	13.379*** (0.990)	6.313*** (1.008)
Second * Rainy	-0.002 (0.016)	-0.019 (0.015)	-2.747 (1.505)	-0.984 (1.342)	-2.923* (1.462)	-2.368 (1.356)
Third * Rainy	-0.006 (0.016)	-0.042** (0.015)	-4.762** (1.527)	-0.497 (1.363)	-2.970* (1.484)	-1.223 (1.377)
Fourth * Rainy	0.000 (0.017)	-0.053*** (0.016)	-7.260*** (1.550)	0.676 (1.390)	-7.907*** (1.506)	-2.582 (1.404)
Fifth * Rainy	0.108*** (0.017)	0.022 (0.016)	-15.634*** (1.579)	-3.146* (1.424)	-7.905*** (1.534)	-0.362 (1.439)
Constant	0.275*** (0.010)	0.077*** (0.011)	52.496*** (0.901)	77.547*** (0.967)	41.525*** (0.876)	60.165*** (0.977)
Observations	27,408	27,408	27,408	27,408	27,408	27,408
R-squared	0.069	0.167	0.146	0.322	0.113	0.239
Mean Dep Var	0.31	0.31	49.34	49.34	32.93	32.93
Controls	Y	Y	Y	Y	Y	Y
Country fixed effects	N	Y	N	Y	N	Y

Notes: Standard error in parentheses. Poorest wealth quintile is omitted from regressions. I assume being surveyed in the rainy season is exogenous to wealth quintiles. Water quality is measured by the levels of E. coli or number of colony forming units (CFUs) per 100mL. Basic services include: piped water, boreholes or tubewells, protected dug wells, protected springs, rainwater, and packaged or delivered water. Limited refers to an improved source more than 30 minutes roundtrip. Unimproved sources include unprotected dug wells and unprotected springs. Surface Water (No service) refers to the direct collection of water from surface waters such as rivers, lakes or irrigation channels.

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Table 4 - Wealth gradient in Water Treatment in the Rainy Season

	Water Treatment		
	(1)	(2)	(3)
Second wealth quintile	-0.021** (0.008)	-0.023** (0.008)	-0.003 (0.006)
Third wealth quintile (Middle)	-0.024** (0.008)	-0.022** (0.008)	0.001 (0.006)
Fourth wealth quintile	0.031*** (0.008)	0.040*** (0.008)	0.030*** (0.007)
Fifth wealth quintile (Richest)	0.012 (0.008)	0.026** (0.009)	0.038*** (0.007)
Rainy season	-0.106*** (0.007)	-0.108*** (0.007)	0.015* (0.006)
Second * Rainy	0.024* (0.011)	0.027* (0.011)	0.005 (0.009)
Third * Rainy	0.038*** (0.011)	0.040*** (0.011)	0.011 (0.009)
Fourth * Rainy	-0.014 (0.011)	-0.011 (0.011)	-0.012 (0.009)
Fifth * Rainy	0.026* (0.012)	0.031** (0.012)	0.001 (0.009)
Constant	0.154*** (0.005)	0.130*** (0.007)	0.024*** (0.006)
Observations	27,408	27,408	27,408
R-squared	0.024	0.037	0.410
Mean Dep Var	0.106	0.106	0.106
Controls	N	Y	Y
Country fixed effects	N	N	Y

Notes: Standard error in parentheses. Basic services include: piped water, boreholes or tubewells, protected dug wells, protected springs, rainwater, and packaged or delivered water. Limited refers to an improved source more than 30 minutes roundtrip. Unimproved sources include unprotected dug wells and unprotected springs. Surface Water (No service) refers to the direct collection of water from surface waters such as rivers, lakes or irrigation channels. I also assume that being surveyed in the rainy season is exogenous to the wealth quintile a household is in. Poorest wealth quintile is excluded from the regressions.

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Table 5 - Wealth gradient in Water Source in the Rainy Season

	Basic water service		Limited water service		Unimproved water service		Surface water	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Second wealth quintile	0.086*** (0.012)	0.081*** (0.012)	0.024** (0.009)	0.023* (0.009)	-0.084*** (0.010)	-0.081*** (0.009)	-0.021** (0.007)	-0.018** (0.007)
Third wealth quintile (Middle)	0.181*** (0.013)	0.143*** (0.012)	0.043*** (0.009)	0.039*** (0.010)	-0.168*** (0.010)	-0.138*** (0.009)	-0.055*** (0.007)	-0.045*** (0.007)
Fourth wealth quintile	0.237*** (0.013)	0.129*** (0.013)	0.108*** (0.010)	0.103*** (0.010)	-0.256*** (0.010)	-0.185*** (0.010)	-0.091*** (0.007)	-0.056*** (0.007)
Fifth wealth quintile (Richest)	0.291*** (0.013)	0.131*** (0.014)	0.165*** (0.010)	0.155*** (0.011)	-0.349*** (0.011)	-0.232*** (0.011)	-0.107*** (0.008)	-0.061*** (0.008)
Rainy season	0.045*** (0.012)	0.005 (0.012)	0.022* (0.009)	0.016 (0.010)	-0.134*** (0.009)	-0.053*** (0.010)	0.072*** (0.007)	0.036*** (0.009)
Second * Rainy	-0.015 (0.018)	-0.013 (0.017)	0.012 (0.013)	0.012 (0.013)	0.044** (0.014)	0.039** (0.013)	-0.044*** (0.010)	-0.042*** (0.009)
Middle * Rainy	-0.043* (0.018)	-0.024 (0.017)	0.014 (0.013)	0.015 (0.013)	0.091*** (0.014)	0.069*** (0.013)	-0.066*** (0.010)	-0.065*** (0.009)
Fourth * Rainy	-0.026 (0.018)	0.014 (0.017)	-0.029* (0.014)	-0.029* (0.014)	0.116*** (0.014)	0.076*** (0.014)	-0.069*** (0.010)	-0.070*** (0.009)
Fifth * Rainy	-0.004 (0.018)	0.048** (0.018)	-0.062*** (0.014)	-0.059*** (0.014)	0.131*** (0.015)	0.073*** (0.014)	-0.071*** (0.010)	-0.068*** (0.009)
Constant	0.398*** (0.008)	0.446*** (0.012)	0.096*** (0.006)	0.057*** (0.010)	0.385*** (0.006)	0.382*** (0.009)	0.110*** (0.005)	0.103*** (0.006)
Observations	27,408	27,408	27,408	27,408	27,408	27,408	27,408	27,408
R-squared	0.046	0.155	0.018	0.032	0.076	0.180	0.047	0.083
Mean Dep Var	0.559	0.559	0.162	0.162	0.195	0.195	0.072	0.072
Controls	N	Y	N	Y	N	Y	N	Y
Country fixed effects	N	Y	N	Y	N	Y	N	Y

Notes: Standard error in parentheses. Basic services include: piped water, boreholes or tubewells, protected dug wells, protected springs, rainwater, and packaged or delivered water. Limited refers to an improved source more than 30 minutes roundtrip. Unimproved sources include unprotected dug wells and unprotected springs. Surface Water (No service) refers to the direct collection of water from surface waters such as rivers, lakes or irrigation channels. I also assume that being surveyed in the rainy season is exogenous to the wealth quintile a household is in. Poorest wealth quintile is excluded from the regressions.

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

APPENDIX B

Table B.1 - Improved Water Variable Conditions

Risk at source	Risk at use	Improved water
Low	Low	1
Low	Mod to High	0
Low	V high	0
Mod to high	Low	1
Mod to high	Mod to High	1 or 0*
Mod to high	V high	0
V high	Low	1
V high	Mod to High	1
V high	V high	0

Notes: * represents that the variable can take both values for that combination of risk levels. If e. coli at source < e. coli at use then it takes the value of 0 and if e. coli at risk level > e. coli at use then takes the value of 1

Table B.2- Description of the Data Sets

Country	Year	HHs tested for water quality	% of HHs surveyed in the rainy season	% of people that treat water before consump.	At source (E. coli per 100 mL)	Point of consump. (E. coli per 100 mL)	<i>% of HH using each water source</i>			
							Basic	Limited	Unimporved	Surface
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(1)	(10)
Benin	2021-22	3965	29.40	0.04	51.692	69.862	0.58	0.13	0.22	0.04
Central African Republic	2018-19	1349	28.16	0.05	45.605	54.351	0.34	0.25	0.33	0.05
Chad	2019	2307	77.85	0.05	57.99	86.05	0.46	0.12	0.27	0.11
DR Congo	2017-18	2783	26.94	0.02	13.11	16.88	19.75	16.8	49.5	12.87
Eswatini	2021-22	1404	83.16	0.08	26.98	35.99	0.68	0.09	0.03	0.12
Gambia	2021-22	1164	12.37	0.08	27.21	35.99	0.68	0.09	0.03	0.12
Ghana	2017-18	3134	59.67	0.03	20.61	41.55	0.63	0.23	0.04	0.07
Guinea Bissau	2018-19	1823	18.05	0.84	37.14	34.49	0.44	0.18	0.36	0.00
Lesotho	2018	1332	15.24	0.05	7.37	17.48	0.77	0.10	0.11	0.00
Madagascar	2018	3264	26.59	0.05	64.52	72.42	0.25	0.12	0.37	0.23
Malawi	2019-20	3121	75.97	0.07	25.21	66.95	0.70	0.18	0.08	0.03
Sierra Leone	2017	1801	86.28	0.07	66.89	71.612	0.43	0.18	0.22	0.17
Togo	2017	1084	57.75	0.10	41.96	60.93	0.59	0.18	0.17	0.06
Zimbabwe	2019	1994	58.02	0.02	26.53	43.02	60.47	19.21	13.03	0.06

Notes: Means of the variables are calculated at the country level.

Table B.3 - Wealth Gradient in the Rainy Season Benin

	Improved Water
Second wealth quintile	0.007 (0.021)
Third wealth quintile (Middle)	0.039 (0.022)
Fourth wealth quintile	0.076*** (0.022)
Fifth wealth quintile (Richest)	0.128*** (0.025)
Rainy season	0.031 (0.025)
Second * Rainy	0.044 (0.038)
Third * Rainy	-0.014 (0.039)
Fourth * Rainy	-0.038 (0.039)
Fifth * Rainy	0.063 (0.039)
HH Head Literate	0.005 (0.013)
Urban	0.014 (0.013)
People in HH	-0.004 (0.002)
Children in HH	0.002 (0.007)
Cosntant	0.096*** (0.018)
R-squared	0.032
Observations	3657

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Standard errors in parenthesis

Table B.4 - Wealth Gradient in the Rainy Season in Central African Republic

	Improved Water
Second wealth quintile	0.041 (0.053)
Third wealth quintile (Middle)	0.001 (0.056)
Fourth wealth quintile	0.042 (0.056)
Fifth wealth quintile (Richest)	0.249*** (0.061)
Rainy season	-0.062 (0.064)
Second * Rainy	0.033 (0.090)
Third * Rainy	0.078 (0.092)
Fourth * Rainy	0.054 (0.091)
Fifth * Rainy	0.137 (0.138)
HH Head Literate	0.018 (0.034)
Urban	0.068 (0.038)
People in HH	-0.004 (0.006)
Children in HH	-0.016 (0.016)
Cosntant	0.225*** (0.048)
R-squared	0.076
Observations	1037

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level
Standard errors in parenthesis

Table B.5 - Wealth Gradient in the Rainy Season in Chad

	Improved Water
Second wealth quintile	0.035 (0.033)
Third wealth quintile (Middle)	0.056 (0.036)
Fourth wealth quintile	0.045 (0.053)
Fifth wealth quintile (Richest)	0.049 (0.036)
Rainy season	0.048 (0.026)
Second * Rainy	-0.059 (0.038)
Third * Rainy	-0.069 (0.041)
Fourth * Rainy	-0.066 (0.056)
Fifth * Rainy	-0.028 (0.037)
HH Head Literate	0.001 (0.012)
Urban	0.034 (0.019)
People in HH	-0.001 (0.002)
Children in HH	-0.005 (0.006)
Cosntant	0.036 (0.024)
R-squared	0.012
Observations	2113

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Standard errors in parenthesis

Table B.6 - Wealth Gradient in the Rainy Season in DR Congo

	Improved Water
Second wealth quintile	0.001 (0.027)
Third wealth quintile (Middle)	0.033 (0.030)
Fourth wealth quintile	0.143*** (0.043)
Fifth wealth quintile (Richest)	0.089 (0.084)
Rainy season	0.027 (0.045)
Second * Rainy	0.004 (0.067)
Third * Rainy	0.110 (0.066)
Fourth * Rainy	0.033 (0.068)
Fifth * Rainy	0.248* (0.098)
HH Head Literate	-0.040 (0.027)
Urban	0.048 (0.029)
People in HH	-0.002 (0.004)
Children in HH	-0.008 (0.011)
Cosntant	0.444*** (0.030)
R-squared	0.052
Observations	2706

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level
Standard errors in parenthesis

Table B.7 - Wealth Gradient in the Rainy Season in Eswatini

	Improved Water
Second wealth quintile	0.036 (0.107)
Third wealth quintile (Middle)	-0.073 (0.102)
Fourth wealth quintile	0.128 (0.111)
Fifth wealth quintile (Richest)	0.323** (0.106)
Rainy season	-0.079 (0.079)
Second * Rainy	0.048 (0.118)
Third * Rainy	0.111 (0.111)
Fourth * Rainy	0.006 (0.120)
Fifth * Rainy	-0.011 (0.115)
HH Head Literate	0.043 (0.041)
Urban	0.106** (0.036)
People in HH	-0.012 (0.006)
Children in HH	-0.007 (0.022)
Cosntant	0.395*** (0.081)
R-squared	0.095
Observations	1164

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Standard errors in parenthesis

Table B.8 - Wealth Gradient in the Rainy Season in Gambia

	Improved Water
Second wealth quintile	0.066 (0.048)
Third wealth quintile (Middle)	0.027 (0.045)
Fourth wealth quintile	0.139** (0.047)
Fifth wealth quintile (Richest)	0.321*** (0.048)
Rainy season	0.017 (0.083)
Second * Rainy	0.084 (0.130)
Third * Rainy	-0.049 (0.123)
Fourth * Rainy	-0.042 (0.133)
Fifth * Rainy	-0.026 (0.132)
HH Head Literate	0.045 (0.041)
Urban	0.097** (0.036)
People in HH	-0.013* (0.006)
Children in HH	-0.006 (0.022)
Cosntant	0.327*** (0.049)
R-squared	0.093
Observations	1164

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Standard errors in parenthesis

Table B.9 - Wealth Gradient in the Rainy Season in Ghana

	Improved Water
Second wealth quintile	0.020 (0.040)
Third wealth quintile (Middle)	0.074 (0.040)
Fourth wealth quintile	0.213*** (0.041)
Fifth wealth quintile (Richest)	0.401*** (0.040)
Rainy season	0.013 (0.031)
Second * Rainy	-0.005 (0.049)
Third * Rainy	-0.019 (0.048)
Fourth * Rainy	-0.051 (0.048)
Fifth * Rainy	-0.038 (0.045)
HH Head Literate	0.005 (0.019)
Urban	0.077*** (0.019)
People in HH	-0.016*** (0.003)
Children in HH	0.000 (0.011)
Cosntant	0.203*** (0.031)
R-squared	0.159
Observations	3134

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Standard errors in parenthesis

Table B.10 - Wealth Gradient in the Rainy Season in Guinea Bissau

	Improved Water
Second wealth quintile	-0.006 (0.035)
Third wealth quintile (Middle)	0.046 (0.038)
Fourth wealth quintile	0.107* (0.042)
Fifth wealth quintile (Richest)	0.131* (0.065)
Rainy season	-0.024 (0.070)
Second * Rainy	-0.026 (0.134)
Third * Rainy	0.068 (0.149)
Fourth * Rainy	-0.120 (0.097)
Fifth * Rainy	-0.236* (0.100)
HH Head Literate	0.014 (0.024)
Urban	-0.029 (0.036)
People in HH	-0.005 (0.004)
Children in HH	-0.002 (0.013)
Cosntant	0.489*** (0.030)
R-squared	0.021
Observations	1823

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Standard errors in parenthesis

Table B.11 - Wealth Gradient in the Rainy Season in Lesotho

	Improved Water
Second wealth quintile	0.138*** (0.040)
Third wealth quintile (Middle)	0.177*** (0.043)
Fourth wealth quintile	0.260*** (0.045)
Fifth wealth quintile (Richest)	0.304*** (0.052)
Rainy season	0.038 (0.129)
Second * Rainy	-0.181 (0.153)
Third * Rainy	-0.148 (0.157)
Fourth * Rainy	-0.163 (0.146)
Fifth * Rainy	-0.178 (0.149)
Urban	0.145*** (0.034)
People in HH	-0.015* (0.006)
Children in HH	-0.014 (0.023)
Cosntant	0.441*** (0.034)
R-squared	0.103
Observations	1332

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Standard errors in parenthesis

Table B.12 - Wealth Gradient in the Rainy Season in Madagascar

	Improved Water
Second wealth quintile	0.042 (0.024)
Third wealth quintile (Middle)	0.051* (0.024)
Fourth wealth quintile	0.083** (0.025)
Fifth wealth quintile (Richest)	0.332*** (0.028)
Rainy season	-0.017 (0.032)
Second * Rainy	-0.005 (0.044)
Third * Rainy	-0.084 (0.046)
Fourth * Rainy	-0.071 (0.047)
Fifth * Rainy	-0.271*** (0.053)
HH Head Literate	0.028 (0.017)
Urban	0.081*** (0.018)
People in HH	-0.001 (0.003)
Children in HH	-0.018 (0.014)
Cosntant	0.119*** (0.025)
R-squared	0.116
Observations	3264

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level
Standard errors in parenthesis

Table B.13 - Wealth Gradient in the Rainy Season in Malawi

	Improved Water
Second wealth quintile	-0.010 (0.041)
Third wealth quintile (Middle)	0.043 (0.040)
Fourth wealth quintile	0.029 (0.039)
Fifth wealth quintile (Richest)	0.145*** (0.040)
Rainy season	-0.056 (0.033)
Second * Rainy	0.016 (0.046)
Third * Rainy	-0.029 (0.045)
Fourth * Rainy	-0.004 (0.044)
Fifth * Rainy	0.015 (0.044)
HH Head literate	-0.008 (0.017)
Urban	0.120*** (0.019)
People in HH	-0.004 (0.003)
Children in HH	-0.004 (0.009)
Cosntant	0.138*** (0.035)
R-squared	0.078
Observations	3121

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Standard errors in parenthesis

Table B.14 - Wealth Gradient in the Rainy Season in Sierra Leone

	Improved Water
Second wealth quintile	0.015 (0.083)
Third wealth quintile (Middle)	0.061 (0.086)
Fourth wealth quintile	0.245* (0.100)
Fifth wealth quintile (Richest)	0.335*** (0.083)
Rainy season	0.037 (0.062)
Second * Rainy	0.038 (0.088)
Third * Rainy	-0.038 (0.091)
Fourth * Rainy	-0.175 (0.101)
Fifth * Rainy	-0.189* (0.084)
HH Head literate	0.021 (0.022)
Urban	-0.044 (0.035)
People in HH	0.009* (0.004)
Children in HH	-0.028* (0.013)
Cosntant	0.106 (0.061)
R-squared	0.029
Observations	1747

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level

Standard errors in parenthesis

Table B.16 - Wealth Gradient in the Rainy Season Togo

	Improved Water
Second wealth quintile	-0.015 (0.053)
Third wealth quintile (Middle)	0.007 (0.054)
Fourth wealth quintile	0.045 (0.066)
Fifth wealth quintile (Richest)	0.222** (0.075)
Rainy season	-0.012 (0.054)
Second * Rainy	-0.022 (0.078)
Third * Rainy	-0.052 (0.062)
Fourth * Rainy	-0.048 (0.080)
Fifth * Rainy	-0.063 (0.085)
HH Head literate	0.035 (0.029)
Urban	0.048 (0.038)
People in HH	-0.001 (0.005)
Children in HH	-0.038* (0.017)
Cosntant	0.182*** (0.044)
R-squared	0.066
Observations	1084

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level
Standard errors in parenthesis

Table B.15 - Wealth Gradient in the Rainy Season in Zimbabwe

	Improved Water
Second wealth quintile	0.010 (0.052)
Third wealth quintile (Middle)	0.005 (0.053)
Fourth wealth quintile	0.075 (0.054)
Fifth wealth quintile (Richest)	0.303*** (0.057)
Rainy season	0.020 (0.047)
Second * Rainy	-0.049 (0.066)
Third * Rainy	-0.016 (0.066)
Fourth * Rainy	-0.056 (0.062)
Fifth * Rainy	-0.088 (0.063)
HH Head literate	0.011 (0.045)
Urban	0.165*** (0.034)
People in HH	-0.017** (0.005)
Children in HH	-0.014 (0.017)
Cosntant	0.256*** (0.056)
R-squared	0.156
Observations	1994

* significant at 10% level; ** significant at 5% level; and *** significant at 1% level
Standard errors in parenthesis

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